

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

Case Study

Code of Conduct:

I Rishiykesh KS (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K.S. Rishiykesh

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CSA07 – COMPUTER NETWORKS

CO3 ASSESSMENT TOOL 2

CASE STUDY

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CASE 1 — REAL-TIME CLOUD VIDEO CONFERENCING

CASE STATEMENT

A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

- 1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.**
- 2. Justify your answer based on the communication requirements of real-time multimedia applications.**
- 3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.**

ABSTRACT

This case examines a real-time multimedia service suffering from packet loss, high jitter and variable bandwidth. The analysis must distinguish the underlying network impairment from the choice of transport protocol and then select improvements that preserve timeliness, continuity and acceptable media quality.

PROBLEM STATEMENT

The central problem is degraded interactive communication caused by 8% packet loss, 180 ms jitter and fluctuating bandwidth. The key analytical task is to determine how transport behavior and application adaptation interact with these conditions.

- 1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.**

Primarily, the quality problem is caused by severe network impairment combined with the application's ability to adapt. UDP is commonly suitable for real-time media because retransmission delay can be harmful, but UDP itself does not create the 8% packet loss or 180 ms jitter. Application buffering and adaptive media control determine how well those conditions are hidden from the user.

KEY REASONING Loss/jitter are network impairments; UDP mainly avoids retransmission delay.

- 2. Justify your answer based on the communication requirements of real-time multimedia applications.**

Real-time multimedia prioritizes timeliness and continuity over perfect delivery. Late packets may be less useful than lost packets, so retransmission-heavy TCP behavior can add delay. A small adaptive jitter buffer smooths arrival variation, while codecs can conceal limited loss.

KEY REASONING Real-time media prioritizes timely delivery over perfect recovery.

3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

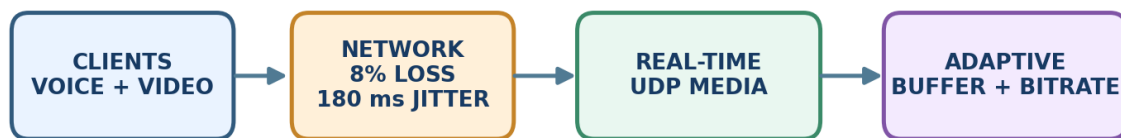
Use a low-latency real-time media transport design, adaptive buffering, adaptive bitrate, loss concealment and QoS. Most importantly, reduce the underlying packet loss and jitter because application tuning cannot fully compensate for a severely impaired network.

KEY REASONING Use a layered fix: network improvement + QoS + adaptive media.

VISUAL ANALYSIS

REAL-TIME CLOUD VIDEO CONFERENCING

NETWORK IMPAIRMENT → REAL-TIME MEDIA RESPONSE



QoS + LOSS CONCEALMENT → lower delay impact → smoother conferencing

The actual communication path and control points

CONCLUSION

The conferencing problem should not be attributed to TCP or UDP alone. The measured loss and jitter are the primary impairments, while the transport and application layers determine how strongly users experience them. A low-delay real-time media approach, adaptive jitter buffering, adaptive bitrate, loss concealment and QoS can improve continuity. However, the most important corrective action is to reduce the underlying packet loss and jitter, because application-level buffering cannot completely compensate for a severely unstable network.

CASE 2 — GLOBAL WEB SERVICES AND DNS

CASE STATEMENT

A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

- 1. Analyze the reasons behind the high DNS resolution time.**
- 2. Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.**
- 3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.**

ABSTRACT

This case focuses on an 800 ms DNS lookup delay in a globally distributed web service. The solution must reduce resolution latency while also ensuring that DNS remains scalable and available when individual servers or regions fail.

PROBLEM STATEMENT

The existing DNS design is producing approximately 800 ms of lookup delay, which postpones the start of application communication. Geographic distance, cache misses, centralized service and DNS record-management choices are potential contributors.

1. Analyze the reasons behind the high DNS resolution time.

An 800 ms DNS lookup creates a large delay before later web communication begins. Geographic distance, cache misses, centralized or overloaded authoritative service, inefficient TTL settings and lack of pre-fetching can all increase lookup time.

KEY REASONING 800 ms shows DNS is a major part of startup delay.

2. Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.

Use Anycast DNS from multiple healthy regions, regional recursive caching, pre-fetching for popular records, optimized TTL values, replicated authoritative servers and health monitoring. The <50 ms value should be treated as a target and validated with regional measurements.

KEY REASONING Anycast shortens paths; caching and pre-fetching reduce repeated work.

3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

Performance improves through shorter paths and cache hits; scalability improves because traffic is distributed; availability improves through replication and health-based routing. The

trade-offs are greater operational complexity, routing/configuration risk and the need to measure the <50 ms target by region and cache state.

KEY REASONING Distribute service and verify the <50 ms target with measurements.

VISUAL ANALYSIS

GLOBAL WEB SERVICES AND DNS

800 ms LOOKUP → DISTRIBUTED RESOLUTION



PRE-FETCH + TTL OPTIMIZATION → fewer repeated lookups → target < 50 ms

The actual communication path and control points

CONCLUSION

A distributed DNS architecture is the appropriate response because the problem is architectural as well as geographic. Anycast can place DNS entry points closer to users, regional caching and pre-fetching can reduce repeated recursive work, and carefully selected TTL values can improve cache efficiency while preserving acceptable freshness. Replicated authoritative servers and health monitoring improve continuity during failures. The below-50-ms target should be validated region by region rather than assumed automatically.

CASE 3 — DISTRIBUTED DNS FOR GLOBAL CLOUD TRAFFIC

CASE STATEMENT

A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly 800 ms because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

- 1. Examine the impact of centralized DNS architecture on application performance.**
- 2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.**
- 3. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.**

ABSTRACT

This case studies the performance and availability problems created when millions of global users depend on a single primary DNS server. The analysis moves from the centralized bottleneck to a distributed design and evaluates how TTL, Anycast and caching affect both speed and resilience.

PROBLEM STATEMENT

The main issue is centralized DNS dependency: all requests reach one primary server, producing nearly 800 ms query time during high-demand events and creating a single point of failure.

1. Examine the impact of centralized DNS architecture on application performance.

A single primary DNS server concentrates geographic distance, processing load and failure risk. During traffic spikes it can become a bottleneck, increasing lookup time. A failure can prevent new clients from resolving the service name.

KEY REASONING One primary creates distance, load concentration and failure risk.

2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.

Use Anycast DNS entry points in multiple regions, regional recursive caches, replicated authoritative DNS, health-based route withdrawal and continuous monitoring of latency, errors, cache-hit ratio and capacity.

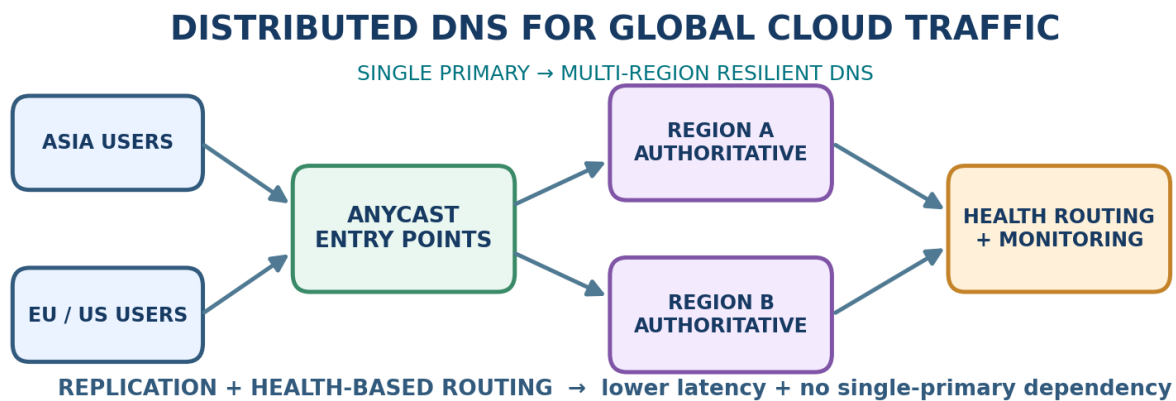
KEY REASONING Each DNS mechanism addresses a different bottleneck.

3. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.

TTL controls how long records remain cached; Anycast distributes users across geographic service points; caching reduces repeated upstream resolution. Together they reduce load and latency while improving resilience, but TTL must balance freshness and cache efficiency.

KEY REASONING Balance speed, freshness and resilience rather than optimizing one alone.

VISUAL ANALYSIS



The actual communication path and control points

CONCLUSION

The single-primary architecture should be replaced by a distributed DNS design with Anycast entry points, regional caching, replicated authoritative servers and health-based routing. This reduces geographic distance, distributes processing demand and limits the impact of a regional or server failure. TTL and caching policies must be selected carefully because aggressive caching can improve speed but delay record changes. Therefore, performance and fault tolerance should be evaluated together rather than optimized independently.

CASE 4 — LOW-LATENCY TCP STOCK TRADING

CASE STATEMENT

A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

- 1. Analyze three possible TCP-related factors responsible for the latency increase, considering flow control, Nagle's algorithm, and SYN queue limitations.**
- 2. Explain how each factor affects transaction processing in high-frequency systems.**
- 3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.**

ABSTRACT

This case examines a latency-sensitive stock-trading platform where acknowledgment delay rises from 1 ms to 40 ms during market opening. The analysis separates three TCP mechanisms—flow control, Nagle's algorithm and SYN queue pressure—and connects each to a distinct source of delay.

PROBLEM STATEMENT

Peak-time transaction latency is increasing alongside retransmissions, slower connection establishment and excessive buffering. The problem therefore requires targeted TCP analysis rather than a single generic configuration change.

- 1. Analyze three possible TCP-related factors responsible for the latency increase, considering flow control, Nagle's algorithm, and SYN queue limitations.**

Flow control can stall senders when receiver windows or buffers are constrained. Nagle's algorithm can delay small writes while waiting to combine data. SYN queue limitations can delay new connections during large bursts of connection attempts. Increased retransmissions also indicate that loss or congestion should be measured.

KEY REASONING Flow control, Nagle and SYN pressure affect different delay stages.

- 2. Explain how each factor affects transaction processing in high-frequency systems.**

Flow-control stalls can increase order queueing and make latency variable. Nagle can add delay to small order or acknowledgment messages. SYN queue pressure can delay the connection itself before an order is transmitted. Retransmissions further increase completion time because lost data must be recovered.

KEY REASONING Small delays become significant in high-frequency trading.

3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

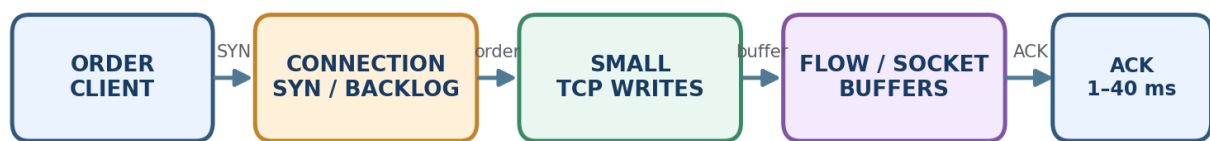
Evaluate TCP_NODELAY for suitable small-message flows, review receive-window and socket-buffer sizing, tune the pending connection/SYN backlog within safe limits, prefer persistent connections where possible, and measure retransmissions, RTT, window utilization and backlog occupancy before and after tuning.

KEY REASONING Tune only measured bottlenecks and validate latency after each change.

VISUAL ANALYSIS

LOW-LATENCY TCP STOCK TRADING

1 ms → 40 ms PEAK LATENCY: WHERE CAN DELAY APPEAR?



TCP_NODELAY + persistent connections + measured buffer/backlog tuning

The actual communication path and control points

CONCLUSION

The latency increase should be addressed through targeted, measured TCP tuning. Flow-control parameters and socket buffers should be reviewed for stalls, TCP_NODELAY should be evaluated for suitable small-message flows, and the SYN backlog should be tuned safely for connection bursts. Persistent connections can also reduce repeated handshake overhead. Because retransmissions are increasing, RTT, loss, window utilization and queue occupancy should be measured before and after changes. This approach improves the probability of lowering latency without sacrificing transaction reliability.